Executive economic summary

Research scenario

INDIVIDUAL GENOMIC ECONOMIC ANALYSIS · HU4AE0F2

PRIMARY MODELED NET MONETARY BENEFIT · under the standardized age-45 scenario ASSUMPTION-DRIVEN SCENARIO

$41,608

MSH2 carrier identified (man)

Primary decision-relevant comparison: Colonoscopy every 2 years to 75 versus no surveillance.

Model basis: hypothetical, built here from published components; its comparator is assumed.

+0.4372	+$2,114	$100,000	Lifetime, to 80
QALYs gained	Incremental cost	Per QALY threshold	Model horizon

5th to 95th percentile of the model draws (parameter uncertainty only): **$36,926** to **$48,861** (see Sensitivity analysis)

MODEL EVIDENCE
Limited: not evidence-supported even at an observed age

DONOR APPLICABILITY
Not established with available context: the donor's age is not observed

Other economic scenarios shown separately, never added

Alternative model: Lynch syndrome carrier identified — **$40,160**
Colonoscopy surveillance after identification · ASSUMPTION-DRIVEN SCENARIO · not counted: it fits this person less well (age at identification)

Alternative model: Lynch syndrome carrier identified (Bennette 2015) — **$90,492**
Return of the finding, then guideline management · ASSUMPTION-DRIVEN SCENARIO · not counted: likely overstates the value (penetrance not corrected for ascertainment)

Cascade testing (relatives): Lynch syndrome carrier identified — **$7,205**
Colonoscopy surveillance after identification · DIFFERENT BENEFICIARY · ASSUMPTION-DRIVEN SCENARIO

SUPPORTED CALCULATION AT ASSUMED AGE 45

None supported — No decision has every model input sourced at the assumed age, so no value is supported there (an empty set, not a measured zero). It is a stricter test than the modeled value above, which rests on stated assumptions; the two are never added.

11 economic results assessed · **4** modeled values · **1** population testing average · **4** true zeros · **0** negative · **1** unquantified · **1** technically unresolved

Values are not additive. Assumed age 45, not observed. A published schedule comparison among known carriers is the secondary comparison, on page 3. Research-genome calls, not clinically confirmed or diagnostic. Exploratory assumption scenarios exist privately and are not part of this report.

Every result, in its evidence class

Every economic result of this genome, grouped by class and ranked within it. Values in different classes are never added. Unknown, unquantified and technically unresolved are never $0. Every finding is a research-genome call, not clinically confirmed.

FINDING (RESEARCH-GENOME CALL) AND DECISION	MODELED NMB	EVIDENCE CLASSIFICATION AND KEY LIMITATION
PRIMARY MODELED 1 · the selected economic model for a decision; it may depend on the stated assumptions shown beside it		
MSH2 carrier identified (man) Colonoscopy every 2 years to 75	**$41,608**	Primary modeled ASSUMPTION-DRIVEN SCENARIO Counted: the selected model's expected value
ALTERNATIVE MODELED 2 · a competing model or sensitivity scenario for the same decision; never added to the primary result		
Lynch syndrome carrier identified Colonoscopy surveillance after identification	**$40,160**	Alternative model ASSUMPTION-DRIVEN SCENARIO Not counted: it fits this person less well (age at identification)
Lynch syndrome carrier identified (Bennette 2015) Return of the finding, then guideline management	**$90,492**	Alternative model ASSUMPTION-DRIVEN SCENARIO Not counted: its penetrance comes from family-based estimates not corrected for ascertainment, so it likely overstates the value to a carrier found by a genome-first screen
FAMILY / CASCADE-TESTING 1 · accrues to relatives or another beneficiary; never in the individual's personal total		
Lynch syndrome carrier identified Colonoscopy surveillance after identification	**$7,205**	Cascade testing (relatives) DIFFERENT BENEFICIARY · ASSUMPTION-DRIVEN SCENARIO Accrues to relatives; a different beneficiary, not added

Results without a figure, each with its reason

POPULATION / TESTING STRATEGY 1 · a value per person screened or tested in a population, not this donor's value; never added

Hypothetical, based on assumptions: any adult, before results — A population testing-strategy average per patient tested, not this genotype's value

TRUE ZERO 4 · the evidence supports no incremental benefit or change in care within the defined decision

ANKK1; DRD2 — ISPG 2019: common psychiatric risk variants are individually of no clinical significance and …

ATP7B rs1061472 (a benign polymorphism, not a Wilson disease finding) — Not a pathogenic variant. ATP7B c.2495A>G p.Lys832Arg (rs1061472), the only variant behind this …

vitamin D (at age 45) — Outside the trial's population at this age: VITAL enrolled men 50 and older and women 55 and …

UNQUANTIFIED 1 · potentially relevant, but the evidence is insufficient for a monetary value; not $0

CBS pathogenic (ClinVar) — No economic model yet

TECHNICALLY UNRESOLVED 1 · the genomic evidence cannot reliably support the interpretation the decision needs; not $0

NAT2 slow acetylator (three-marker inference): NAT2-guided isoniazid dosing — NAT2 is inferred here from three markers (*5/*6/*7) without phase

Research scenario

What changes because of the genetic finding

PRIMARY DECISION-RELEVANT COMPARISON

WITHOUT THE GENOMIC FINDING

No surveillance

Assumed comparator: An unidentified carrier would receive average-risk screening, so surveillance against none overstates the increment

WITH THE GENOMIC FINDING

Colonoscopy every 2 years to 75

+0.43722	×	$100,000	−	$2,114	=	$41,608
QALYs gained		per QALY		incremental cost		net monetary benefit

ASSUMPTION-DRIVEN SCENARIO The identity reproduces the printed value from its printed terms (costs in 2019 US dollars; positive is more spending). Cost per QALY gained, calculated from the two terms above: **$4,835** per QALY gained, below the $100,000 threshold.

Population: a man with an incidental MSH2 finding · **Valued per:** carrier identified · **Horizon:** lifetime, to 80 · **Perspective:** US, health-care sector (Medicare-based costs) · **Discounting:** 3% a year · **Price year:** source 2019, restated to 2019 USD · **Sources:** GenomeLens model (hypothetical), built here from published components (PMID 35809086; PMID 33839100)

WHO EACH VALUE ACCRUES TO

TO THE PERSON
+0.4372 QALYs
NMB $41,608
ASSUMPTION-DRIVEN SCENARIO

COST TO HEALTH CARE
+$2,114
In the source model (perspective: US, health-care sector), 2019 USD
ASSUMPTION-DRIVEN SCENARIO

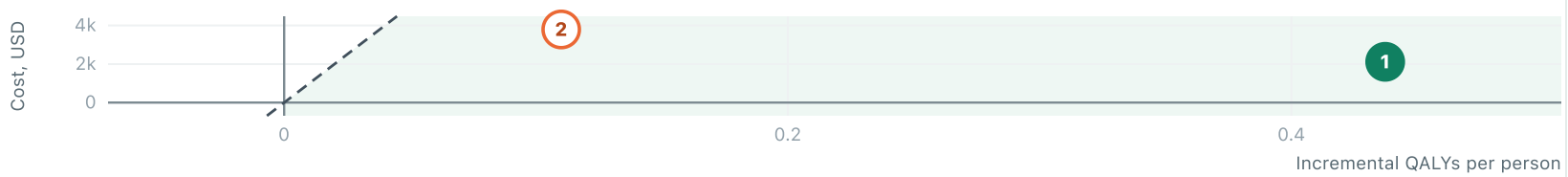
TO RELATIVES
$7,205
Cascade testing, NMB
DIFFERENT BENEFICIARY · ASSUMPTION-DRIVEN SCENARIO

TO THE POPULATION
Not shown here
A per-person-tested population average is a different question; it is listed on page 2 without a figure.

Each tile accrues to a different beneficiary, as the model assigns it; the four are never added.

Where each modeled decision falls

FIGURE 1



primary modeled · cascade testing (relatives) · - - - $100,000 per QALY: below the line and right of the axis is cost-effective at that threshold

	TABLE 1. THE PLOTTED VALUES	ΔQALY	ΔCOST	NMB
1	**Primary modeled**: MSH2 carrier identified (man) Colonoscopy every 2 years to 75 · ASSUMPTION-DRIVEN SCENARIO	**+0.4372**	**+$2,114**	**$41,608**
2	**Cascade testing (relatives)**: Lynch syndrome carrier identified Colonoscopy surveillance after identification · DIFFERENT BENEFICIARY · ASSUMPTION-DRIVEN SCENARIO	**+0.11**	**+$3,795**	**$7,205**

Each point is a different model or beneficiary (own comparator, population, horizon); no two are added. 2 further values are listed on page 2.

State: the figures on this page are model outputs, labeled UNRESOLVED (not established for this donor) unless a row carries another label. Standardized scenario at assumed age 45: the donor-specific economic value is not established with available context.

Research scenario

How robust the modeled value is

Three questions: how wide the parameter uncertainty is, whether another published model would change the value, and how much it depends on the willingness-to-pay threshold. Each answer is a stored model output; nothing is re-estimated here.

1 · PARAMETER UNCERTAINTY

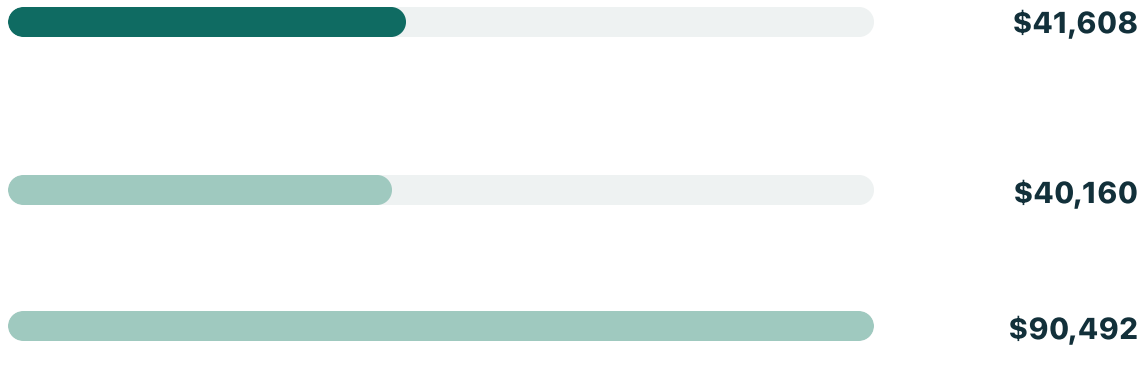
Primary modeled NMB distribution

FIGURE 2



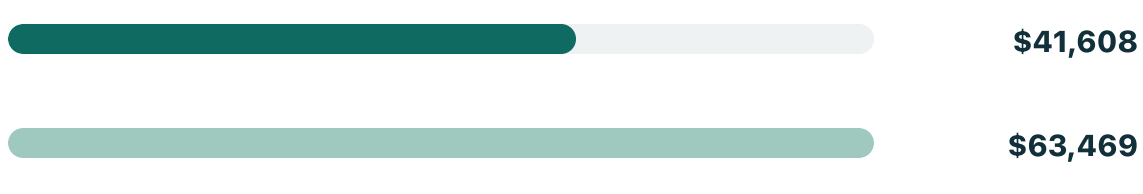
ASSUMPTION-DRIVEN SCENARIO Reported model draws (ranges the sources print or the registry holds, triangular, independent; parameter uncertainty only); the interval is not necessarily a confidence interval for real-world effectiveness.

2 · MODEL CHOICE (STRUCTURAL UNCERTAINTY) · FIGURE 3



Why each was not selected: Guzauskas 2022: it fits this person less well (age at identification); Bennette 2015: likely overstates the value (penetrance not corrected for ascertainment). Alternatives are never added.

3 · WILLINGNESS-TO-PAY THRESHOLD · FIGURE 4



Calculated from the printed terms: 0.43722 QALYs × $150,000 − $2,114. It is a sensitivity case, not a second value, and the two are never added.

4 · THE AGE ASSUMPTION

Recomputed at the stated age. The value is recomputed at age 45 from the published inputs; the source's own result is for another age. The age is stated for this run, not observed.

Research scenario

What the value rests on, and what it cannot claim

CLINICAL VALIDITY OF THE GENOMIC CALLS

Research-genome calls. Genomic calls in this proof of concept have not been independently confirmed by a clinical laboratory and are not diagnostic results. Before any clinical use, each finding would need confirmation by a clinical laboratory. The modeled values assume the call is correct.

- **NAT2 slow acetylator (three-marker inference): NAT2-guided isoniazid dosing** — technically unresolved: the genomic evidence cannot support the interpretation

EVIDENCE SUMMARY

CLASS	RESULTS	WHAT THE CLASS MEANS
Primary modeled	**1**	the selected economic model for a decision; it may depend on the stated assumptions shown beside it
Alternative model	**2**	a competing model or sensitivity scenario for the same decision; never added to the primary result
Cascade testing (relatives)	**1**	accrues to relatives or another beneficiary; never in the individual's personal total
Testing strategy (population)	**1**	a value per person screened or tested in a population, not this donor's value; never added
True zero	**4**	the evidence supports no incremental benefit or change in care within the defined decision
Unquantified	**1**	potentially relevant, but the evidence is insufficient for a monetary value; not $0
Technically unresolved	**1**	the genomic evidence cannot reliably support the interpretation the decision needs; not $0

The primary value's model evidence is limited: it is not evidence-supported even at an observed age. Its inputs are drawn from the published sources cited on page 6 and from the stated assumptions below, not estimated from this genome; the inputs, their values and their evidence grading are not published.

ASSUMPTIONS

- Standardized age 45, stated for this run (one assumed age for every demonstration genome), not observed
- Willingness to pay $100,000 per QALY; costs and QALYs discounted at 3% a year unless a source states its own rate
- Costs restated to 2019 US dollars with the medical-care consumer price index
- Each decision is valued once; values in different classes are never added
- The no-surveillance comparator is an assumption: an unidentified carrier would receive average-risk screening, so surveillance against none overstates the increment. The model's death-only crediting pulls the other way; neither bias is sized.
- At the stated age of 45, assumed for this run, not observed
- If identified at 45
- No surveillance: an assumption of this model, not the source's comparator arm
- Comparator: no surveillance; age stated, not observed
- The 2-yearly effect (hazard ratio 0.535) is the source's assumption: 0.05 below its fitted 3-yearly 0.585

LIMITATIONS

- The assessment is incomplete: 1 unquantified and 1 technically unresolved results carry no figure; that is not a conclusion that the genome has no further value.
- The supported calculation at the assumed age is none supported; the modeled values rest on stated assumptions.
- No model here has been externally validated. The published alternative of Bennette 2015 is 2.2 times the counted value and is not counted; neither candidate model is clean. Exploratory assumption scenarios exist privately and are not part of this report.

Research scenario

Where every figure comes from

SOURCES BEHIND EVERY MODELED VALUE

VALUE	FIGURE	SOURCE (PUBLISHED CITATIONS)	PRICE BASIS
Primary modeled MSH2 carrier identified (man)	**$41,608** ASSUMPTION-DRIVEN SCENARIO	GenomeLens model (hypothetical), built here from published components (PMID 35809086; PMID 33839100)	source 2019 prices, restated to 2019 USD
Alternative model Lynch syndrome carrier identified	**$40,160** ASSUMPTION-DRIVEN SCENARIO	Guzauskas 2022 (published model, restated) (PMID 35227606)	source 2021 prices, restated to 2019 USD
Alternative model Lynch syndrome carrier identified (Bennette 2015)	**$90,492** ASSUMPTION-DRIVEN SCENARIO	Bennette 2015 (published model, restated) (PMID 25394171)	source 2013 prices, restated to 2019 USD
Cascade testing (relatives) Lynch syndrome carrier identified	**$7,205** DIFFERENT BENEFICIARY · ASSUMPTION-DRIVEN SCENARIO	PMID 35227606	source 2021 prices, restated to 2019 USD

WHAT WAS CHECKED AGAINST PUBLISHED SOURCES

1 published table recomputed from its printed totals

1 model reproduces its source's printed outputs

1 model cross-checked against a model by others

0 models externally validated against observed outcomes

These are technical checks against published sources, not validation of the models: none of the decision models has been validated against observed outcomes. The figures behind the checks are in the public reproduction kit and in `docs/VALIDATION.md`.

DATA PROVENANCE

INPUT	PGP-HMS sample hu4AE0F2, a public research genome
CALLS	research-genome calls, not confirmed by a clinical laboratory
GENERATED	2026-10-24

WHY THE FIGURES CAN BE RECOMPUTED

- ✓ Every modeled value keeps its terms (the QALY change, the cost change and the threshold), so its arithmetic can be recomputed independently; page 3 shows the identity for the primary value.
- ✓ Each modeled value names its published sources, and each assumption is named.
- ✓ Alternative models, values to relatives and conditional values are kept apart and never added; a result that cannot be priced is classed with its reason: unknown is not $0.

REPORTING STANDARD AND WHAT IS NOT PUBLISHED

The public HEOR technical dossier (demonstration 1) maps every CHEERS 2022 item (12 of 28 addressed, 8 in part, 8 not reported; not a claim of compliance). The model's inputs, structure and registry are proprietary and are not published, nor are the private editions. This report is a separate public export that prints a fixed list of fields: every decimal, and every whole number of 100 or more apart from years, also appears in this genome's public reports, the public documents or the reproduction kit's recorded output, or is calculated here from printed terms.

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